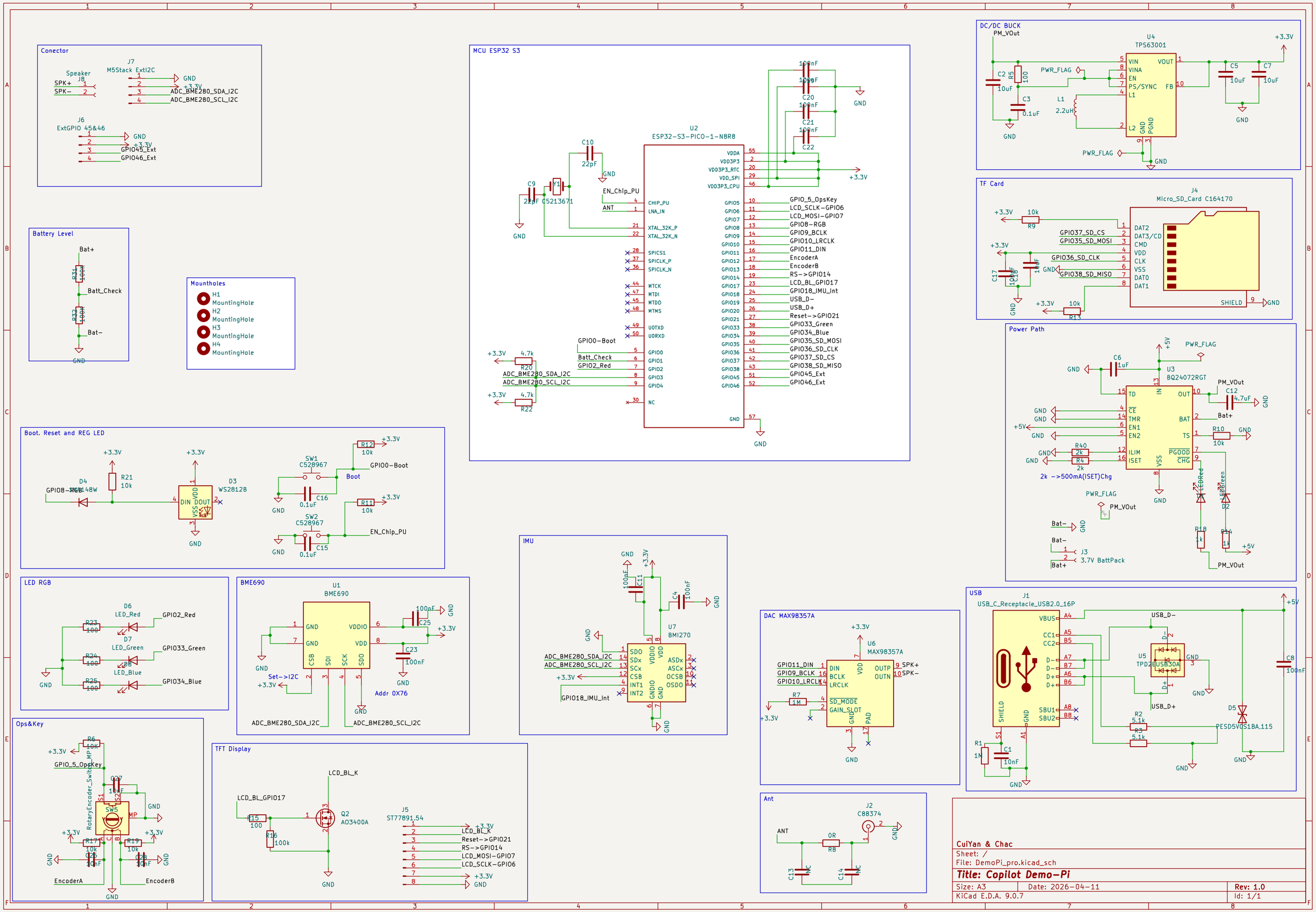
Conector
J7
M5Stack ExtI2C
Speaker
J8
SPK+
SPK−
GND
+3.3V
ADC_BME280_SDA_I2C
ADC_BME280_SCL_I2C
J6
ExtGPIO 45&46
GPIO45_Ext
GPIO46_Ext
MCU ESP32 S3
U2
ESP32-S3-PICO-1-N8R8
C10
22pF
C9
22pF
Y1
C5213671
EN_Chip_PU
ANT
CHIP_PU
LNA_IN
XTAL_32K_P
XTAL_32K_N
SPICS1
SPICLK_P
SPICLK_N
MTCK
MTDI
MTDO
MTMS
U0TXD
U0RXD
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
NC
VDDA
VDD3P3
VDD3P3_RTC
VDD_SPI
VDD3P3_CPU
C20
100nF
C21
100nF
C22
GPIO0−Boot
Batt_Check
GPIO2_Red
R20
4.7k
R22
4.7k
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO17
GPIO18
GPIO19
GPIO20
GPIO21
GPIO33
GPIO34
GPIO35
GPIO36
GPIO37
GPIO38
GPIO45
GPIO46
GPIO_5_OpsKey
LCD_SCLK−GPIO6
LCD_MOSI−GPIO7
GPIO8−RGB
GPIO9_BCLK
GPIO10_LRCLK
GPIO11_DIN
EncoderA
EncoderB
RS−>GPIO14
LCD_BL_GPIO17
GPIO18_IMU_Int
USB_D−
USB_D+
Reset−>GPIO21
GPIO33_Green
GPIO34_Blue
GPIO35_SD_MOSI
GPIO36_SD_CLK
GPIO37_SD_CS
GPIO38_SD_MISO
GPIO45_Ext
GPIO46_Ext
DC/DC BUCK
PM_VOut
U4
TPS63001
VIN
VINA
EN
PS/SYNC
L1
L2
GND
PGND
VOUT
FB
C2
10uF
R5
100
C3
0.1uF
L1
2.2uH
PWR_FLAG
C5
10uF
C7
10uF
+3.3V
TF Card
J4
Micro_SD_Card C164170
DAT2
DAT3/CD
CMD
VDD
CLK
VSS
DAT0
DAT1
SHIELD
GPIO37_SD_CS
GPIO35_SD_MOSI
GPIO36_SD_CLK
GPIO38_SD_MISO
R9
10k
R13
10k
C17
100nF
Battery Level
Bat+
R31
100k
Batt_Check
R32
100k
Bat−
Mounthole
H1 MountingHole
H2 MountingHole
H3 MountingHole
H4 MountingHole
Power Path
U3
BQ24072RGT
TD
CE
TMR
EN1
EN2
ILIM
ISET
IN
OUT
BAT
TS
PGOOD
CHG
VSS
C6
1uF
+5V
PM_VOut
C12
4.7uF
Bat+
R10
10k
R40
2k
R4
2k
2k −>500mA(ISET)Chg
LED Red
LED Green
D2
R18
1k
R14
1k
Bat−
J3
3.7V BattPack
Boot, Reset and REG LED
D4
WS2814BW
R21
10k
D3
WS2812B
DIN
DOUT
VDD
VSS
SW1
C528967
SW2
C528967
C16
0.1uF
C15
0.1uF
R12
10k
R11
10k
Boot
GPIO0−Boot
EN_Chip_PU
LED RGB
D6
LED_Red
D7
LED_Green
D8
LED_Blue
R23
100
R24
100
R25
100
GPIO2_Red
GPIO33_Green
GPIO34_Blue
BME690
U1
BME690
GND
VDDIO
VDD
CSB
SDI
SCK
SDO
C25
100nF
C23
100nF
Set−>I2C
Addr 0X76
ADC_BME280_SDA_I2C
ADC_BME280_SCL_I2C
IMU
U7
BMI270
SDO
SDx
SCx
CSB
INT1
INT2
GNDIO
VDDIO
VDD
ASDx
ASCx
OCSB
OSDO
C11
100nF
C4
100nF
GPIO18_IMU_Int
DAC MAX98357A
U6
MAX98357A
DIN
BCLK
LRCLK
SD_MODE
GAIN_SLOT
OUTP
OUTN
PAD
GPIO11_DIN
GPIO9_BCLK
GPIO10_LRCLK
R7
1M
SPK+
SPK−
USB
J1
USB_C_Receptacle_USB2.0_16P
VBUS
CC1
CC2
D−
D+
SBU1
SBU2
SHIELD
U5
TPD2EUSB30A
D5
PESD5V0S1BA,115
C8
100nF
R2
5.1k
R3
5.1k
R1
1M
C1
10nF
Ops&Key
R6
10k
GPIO_5_OpsKey
RotaryEncoder_Switch_MP
SW5
C27
100nF
R17
10k
R19
10k
C26
10nF
C28
10nF
EncoderA
EncoderB
TFT Display
LCD_BL_K
LCD_BL_GPIO17
R15
100
R16
100k
Q2
AO3400A
J5
ST77891.54
LCD_BL_K
Reset−>GPIO21
RS−>GPIO14
LCD_MOSI−GPIO7
LCD_SCLK−GPIO6
Ant
J2
C88374
ANT
0R
R8
C13
NC
C14
NC
CuiYan & Chac
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Title: Copilot Demo−Pi
Size: A3
Date: 2026−04−11
Rev: 1.0
KiCad E.D.A. 9.0.7
Id: 1/1